

# Tutorial\_2 list of question and answer

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## 1. Features of OFDMA (Orthogonal Frequency Division Multiple Access)

OFDMA is the **key multiple access technique** used in **WiMAX and LTE**. It's basically an advanced version of **OFDM (Orthogonal Frequency Division Multiplexing)** where resources are shared among many users at the same time.

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### ✨ Main Features (in simple terms):

#### 1. Multiple Users at Once

- OFDMA divides the available frequency band into many **small subcarriers**.
  - Each user is given a different set of subcarriers → so **many users can transmit at the same time** without interfering.
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#### 2. Orthogonality = No Overlap Interference

- The subcarriers are placed **orthogonally** (mathematically aligned).
  - This ensures they don't interfere with each other, even though they are tightly packed.
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#### 3. Efficient Bandwidth Use

- Unlike old systems where bandwidth was wasted to prevent overlap, OFDMA uses **every bit of spectrum efficiently**.
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#### 4. Robust(মজবুত) Against Multipath Fading

- In wireless, signals bounce off buildings/objects and overlap.
  - OFDMA handles this well by splitting the data into small chunks and sending them on different subcarriers.
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## 5. Adaptive Resource Allocation

- The system can **dynamically assign subcarriers** depending on channel conditions.
- Example: A user in a weak signal area can get **more subcarriers** for reliability.

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## 6. Supports High Data Rate

- Because multiple narrowband subcarriers are used in parallel, OFDMA can transmit **a large amount of data very fast**.

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## 7. Flexible Uplink & Downlink

- Works efficiently for both **downlink (from base station to user)** and **uplink (from user to base station)**.

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## 8. Good for Low-Power Devices

- Users don't need to use the whole bandwidth, just their subcarriers → saves power (important for mobile devices).



## Quick Recap Table

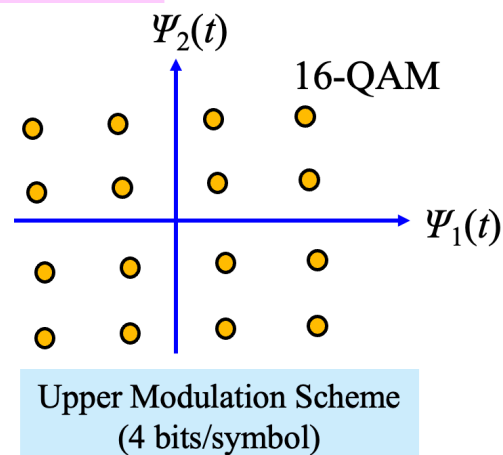
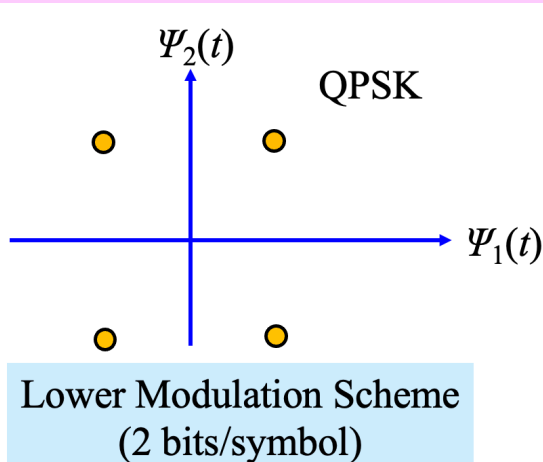
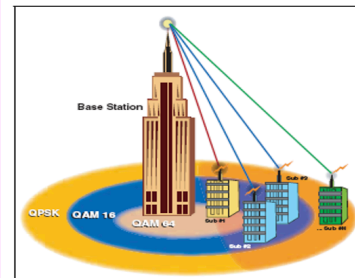
| Feature                 | Easy Meaning                        |
|-------------------------|-------------------------------------|
| Multi-user Access       | Many users share bandwidth at once  |
| Orthogonality           | No interference between subcarriers |
| Bandwidth Efficiency    | Uses spectrum fully                 |
| Multipath Robustness    | Handles reflected signals well      |
| Adaptive Allocation     | Dynamically assigns subcarriers     |
| High Data Rate          | Supports fast data transmission     |
| Uplink/Downlink Support | Works in both directions            |
| Low Power Use           | Saves energy for mobile devices     |

## 2. Adaptive Modulation and Coding (AMC)

### What it is:

Adaptive Modulation and Coding (AMC) is a smart technique used in **4G systems (WiMAX, LTE)** where the **modulation type** and **coding rate** are **automatically adjusted** based on the quality of the wireless channel.

**Adaptive Modulation and Coding:** The IEEE 802.16e standard changes **modulation** and **channel coding** scheme based on received SNR. For example a SS (Subscriber Station) close to the BS can use a high modulation scheme (more bits per symbol) i.e. the system can get more capacity but when the SS is at the cell boarder the system permits lower modulation scheme (increased signal space on orthogonal basis function coordinate system) to avoid huge symbol error rate.



8

## How it works (Step by Step):

### 1. Channel Condition Checked

- The system measures how strong/weak the wireless signal is (using SNR – Signal-to-Noise Ratio).

### 2. If Channel is Good (strong signal)

- Use **higher-order modulation** (like 64-QAM).
- More bits per symbol → **higher data rate**.
- Example: Watching HD video needs this.

### 3. If Channel is Poor (weak or noisy signal)

- Use **lower-order modulation** (like QPSK).

- Fewer bits per symbol → **lower data rate but more reliable**.
- Example: During bad weather or far from tower.

#### 4. Coding Rate Adjusted

- Coding adds **error correction**.
- In bad channels → **more coding bits** (strong error protection).
- In good channels → **less coding bits** (saves bandwidth).

### 🌟 Key Features of AMC:

- Dynamically changes modulation (QPSK, 16-QAM, 64-QAM).
- Adjusts coding for **error protection**.
- Ensures **best possible data rate** under given conditions.
- Improves **spectrum efficiency** and **user experience**.

### 🧠 Simple Analogy:

Imagine talking to a friend:

- If the line is **clear**, you speak **fast** and **use complex words** (high modulation, less coding).
  - If the line is **noisy**, you **slow down** and use **simpler words** (low modulation, more coding).
- ✅ That's exactly what AMC does in wireless communication.

### 📊 Quick Recap Table

| Channel Condition | Modulation Used | Coding Rate | Data Rate | Reliability |
|-------------------|-----------------|-------------|-----------|-------------|
| Good (High SNR)   | 64-QAM          | Low coding  | High      | Medium      |
| Medium            | 16-QAM          | Medium      | Medium    | Medium      |
| Poor (Low SNR)    | QPSK            | High coding | Low       | High        |

### 3. 🔄 TDD vs FDD



Both TDD and FDD are **duplexing techniques** used in wireless systems (like LTE, WiMAX) to allow **uplink (user → base station)** and **downlink (base station → user)** communication.

## Comparison Table

| Feature                    | TDD (Time Division Duplex)                                                                 | FDD (Frequency Division Duplex)                                                |
|----------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Basic Idea</b>          | Uplink & downlink share the <b>same frequency band</b> but use <b>different time slots</b> | Uplink & downlink use <b>different frequency bands</b> at the <b>same time</b> |
| <b>Spectrum Use</b>        | One frequency band is enough (more spectrum efficient)                                     | Needs two separate frequency bands (one for uplink, one for downlink)          |
| <b>Cost</b>                | Cheaper (uses single band, simpler hardware)                                               | More expensive (needs two bands + duplexer hardware)                           |
| <b>Traffic Flexibility</b> | Can dynamically adjust time slots (good for asymmetric traffic like internet browsing)     | Fixed allocation (good for symmetric traffic like voice calls)                 |
| <b>Synchronization</b>     | Needs strict time synchronization between users                                            | No strict time sync needed (separate bands avoid clash)                        |
| <b>Example Use</b>         | WiMAX, TD-LTE, some 4G/5G systems                                                          | GSM, WCDMA, LTE-FDD                                                            |
| Latency                    | Higher latency due to switching                                                            | lower latency due to continuous transmitting                                   |
| communication              | Half-duplex on the same channel                                                            | Full-duplex with paired channel                                                |

## Simple Analogy:

- **TDD:** Like one road used by cars in **both directions**, but cars take turns (time slots).
- **FDD:** Like having **two separate roads** — one for each direction (uplink & downlink at the same time).

## Quick Summary

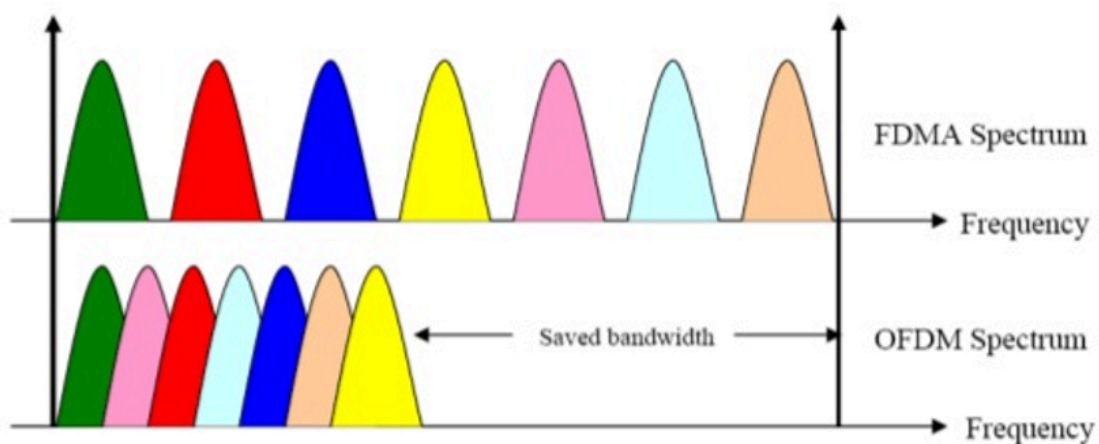
- **TDD = Time sharing, single band, cheaper, flexible.**

- FDD = Frequency splitting, two bands, costlier, good for traditional voice.

## 4. 📡 Why OFDMA is Preferable in Wireless Networks?

1 question answer cover this question

OFDMA (**Orthogonal Frequency Division Multiple Access**) is used in modern wireless systems like **LTE** and **WiMAX**, and it's preferred because it solves many problems of older techniques.



### ✨ Reasons why OFDMA is better:

#### 1. Supports Multiple Users at the Same Time

- OFDMA splits the channel into many **small subcarriers**.
- Different users can be given different sets of subcarriers → so many users share the same frequency at once without interfering.

#### 2. Efficient Spectrum Use

- Subcarriers are placed **orthogonally** (mathematically aligned).
- This prevents overlap and makes **full use of available bandwidth**.

#### 3. High Data Rate

- Since many subcarriers work in parallel, large amounts of data can be transmitted quickly.
- Perfect for **video streaming, gaming, and high-speed internet**.

#### 4. Handles Multipath Fading

- Wireless signals often reflect off buildings/objects and interfere.
  - OFDMA divides data into smaller pieces over different subcarriers → making it **resistant to interference and fading**.
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#### 5. Adaptive to Channel Conditions

- Subcarriers can be dynamically assigned based on signal strength.
  - Example: A user far from the tower may get **stronger modulation with fewer errors**, while a nearby user gets **higher-order modulation for speed**.
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#### 6. Low Power Requirement

- Users don't have to transmit on the entire bandwidth, just their assigned subcarriers → saves **battery life** in mobile devices.
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#### 7. Flexibility for Asymmetric Traffic

- In real internet use, **download > upload**.
  - OFDMA allows flexible resource allocation (more subcarriers for download, fewer for upload).
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### **In short:**

OFDMA is **preferable** because it:

- ✓ Supports multiple users efficiently
  - ✓ Uses spectrum fully
  - ✓ Handles interference well
  - ✓ Gives higher data rates
  - ✓ Saves power for devices
- 

### **Analogy:**

Think of a highway:

- **Old methods:** One car (user) takes the whole highway.

- **OFDMA:** The highway is divided into many lanes (subcarriers). Each car (user) gets its own lane, so many cars can move smoothly at once. 🚗🚙🚐

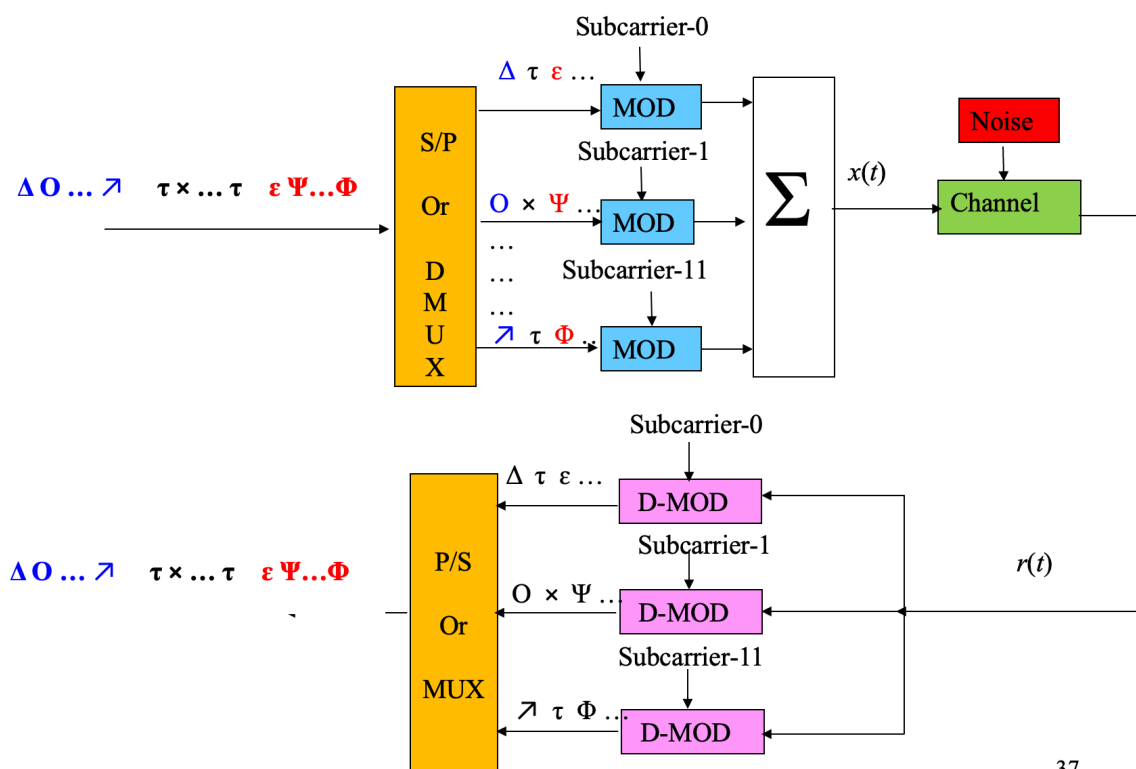
## 5. 📡 Explain OFDMA-Based Physical Layer

OFDMA: The access technique used in physical layer of WiMAX is OFDMA. where the high speed serial data is converted to low rate parallel streams and each stream is modulated by separate carrier each one is known as subcarrier. Subcarriers are mutually orthogonal and deals with low rate data rate hence can protect ISI resulting from multipath propagation.

The **physical layer** is the bottom layer of a wireless system – it deals with **how bits are actually transmitted over the air**.

In WiMAX/LTE, the physical layer uses **OFDMA (Orthogonal Frequency Division Multiple Access)** as its foundation.

## OFDMA-based Physical Layer Access



37

## **How it Works (Step by Step):**

### 1. Data Input

- The user data (bits) enter the physical layer.

### 2. Channel Coding & Interleaving

- Data is encoded (error correction) and shuffled (interleaving) to protect against errors due to noise and fading.

### 3. Modulation Mapping

- Data bits are mapped onto symbols using modulation techniques like **QPSK, 16-QAM, or 64-QAM** depending on channel condition (Adaptive Modulation & Coding).

### 4. Subcarrier Allocation

- The modulated symbols are placed onto different **orthogonal subcarriers**.
- Each user gets its own set of subcarriers.

### 5. IFFT (Inverse Fast Fourier Transform)

- Converts frequency-domain subcarriers into a single **time-domain OFDM signal** for transmission.

### 6. Cyclic Prefix Insertion

- A guard interval is added between OFDM symbols to fight multipath interference (signal reflections).

### 7. Transmission over Air

- The OFDM signal is transmitted through the antenna over the wireless channel.

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## **At the Receiver (Reverse Process):**

- The receiver removes cyclic prefix, applies **FFT**, extracts subcarriers, demodulates symbols, corrects errors, and finally recovers the original data bits.

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## **Key Features of OFDMA Physical Layer:**

- Divides bandwidth into many orthogonal subcarriers.

- Allocates subcarriers to multiple users simultaneously.
- Supports **adaptive modulation and coding (AMC)**.
- Handles multipath fading effectively.
- Provides **high data rates and efficient spectrum use**.

## **Analogy:**

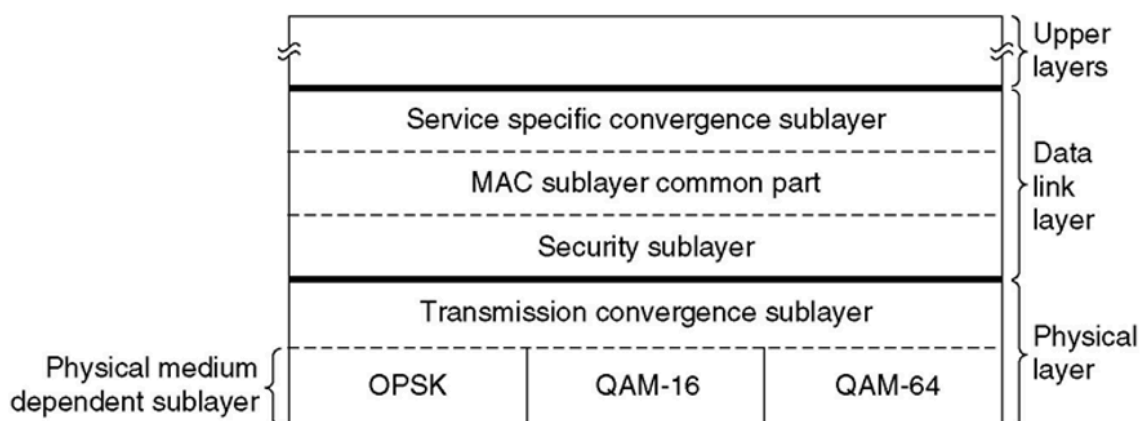
Think of OFDMA physical layer as a **music concert hall**:

- Each singer (user) sings in their **own pitch (subcarrier)**.
- The audience (receiver) can separate each singer because their voices are **orthogonal** (non-overlapping).
- Together, it sounds harmonious without interference. 🎵

## 6. **Explain OFDMA-Based Data Link Layer**

### Data link layer

From the reference model as illustrated in Figure 5, there are three sub-layers in the data link layer composed of i) a **security sub-layer**, ii) a **MAC common part sub-layer**, and iii) a **convergence sub-layer**. It provides only connection oriented service



The **data link layer** in WiMAX/LTE using OFDMA is responsible for **how users share the wireless channel fairly and efficiently**.

It has **two sublayers**:

1. **MAC (Medium Access Control) Sublayer**
  2. **Logical Link Control (LLC) Sublayer**
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## 1. MAC Sublayer (Key Role in OFDMA)

- The **MAC scheduler** at the base station (BS) decides **who gets which subcarriers and for how long**.
- Divides resources into **time slots × frequency subcarriers** → forms a **time–frequency grid**.

### Main Functions of MAC in OFDMA:

- **Resource Allocation:** Assigns subcarriers to users dynamically.
  - **QoS (Quality of Service):** Ensures priority for real-time traffic (like video calls) and fairness for others.
  - **Error Handling:** ARQ (Automatic Repeat Request) used for retransmission if errors occur.
  - **Scheduling:** Different scheduling algorithms (Round Robin, Proportional Fair, etc.) used.
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## 2. LLC Sublayer

- Provides an interface between the **network layer (IP)** and the **MAC sublayer**.
  - Ensures **error control and flow control**.
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## How OFDMA Works at Data Link Layer (Step by Step):

1. User data enters from the **network layer**.
2. The **LLC sublayer** ensures proper error detection and flow control.
3. The **MAC scheduler**:
  - Checks channel condition of each user.
  - Allocates **subcarriers + time slots** based on demand and QoS.
  - Example: A video call user gets more stable slots, while a file download user can wait a bit.

4. The assigned data goes down to the **physical layer**, where it's mapped onto subcarriers and transmitted.
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### ✨ **Key Features of OFDMA Data Link Layer:**

- **Dynamic Resource Scheduling** (based on user demand & channel quality).
  - **QoS Support** (real-time vs non-real-time).
  - **Error Control** (retransmission, ARQ).
  - **Efficient Use of Resources** (multiuser sharing of bandwidth).
  - **Fairness** among users.
- 

### 🧠 **Simple Analogy:**

Think of the data link layer like an **exam invigilator (MAC scheduler)** in a hall:

- Students (users) are assigned **seats (subcarriers + time slots)**.
  - The invigilator ensures **no two students sit on the same seat (no collision)**.
  - Priority students (disabled or VIP) get better seats (QoS).
  - If a student makes a mistake (error), they're given another chance (retransmission).
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### ✅ **In short:**

- **Physical Layer** → handles how data is transmitted on subcarriers.
  - **Data Link Layer** → decides **which user gets which subcarriers and when**, ensuring fairness, QoS, and efficiency.
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## 7. 📦 **Draw the MAC PDU Frame Format (WiMAX style)**



- ✓ Each MAC frame is prefixed with GMH (generic MAC header).
- ✓ Field (in SH) to indicate whether the payload is encrypted or not. If the payload is encrypted then the encryption key is also given.
- ✓ Header CRC field is a checksum over the header only using the generator polynomial  $x^8+x^2+x+1$ . The length of this field is 8bits.

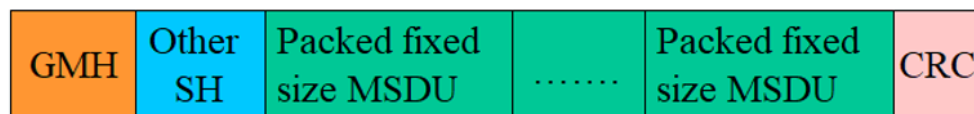


Fig.6 MAC PDU frame carrying several-fixed length MSDUs packed together

GMH → Generic MAC Header (used for carrying data and MAC-layer signaling messages)

SH → Sub-header



Fig.7 MAC PDU frame carrying a single fragmented MSDU

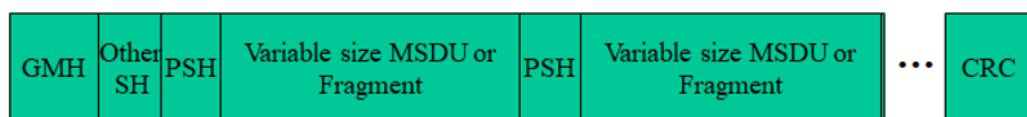
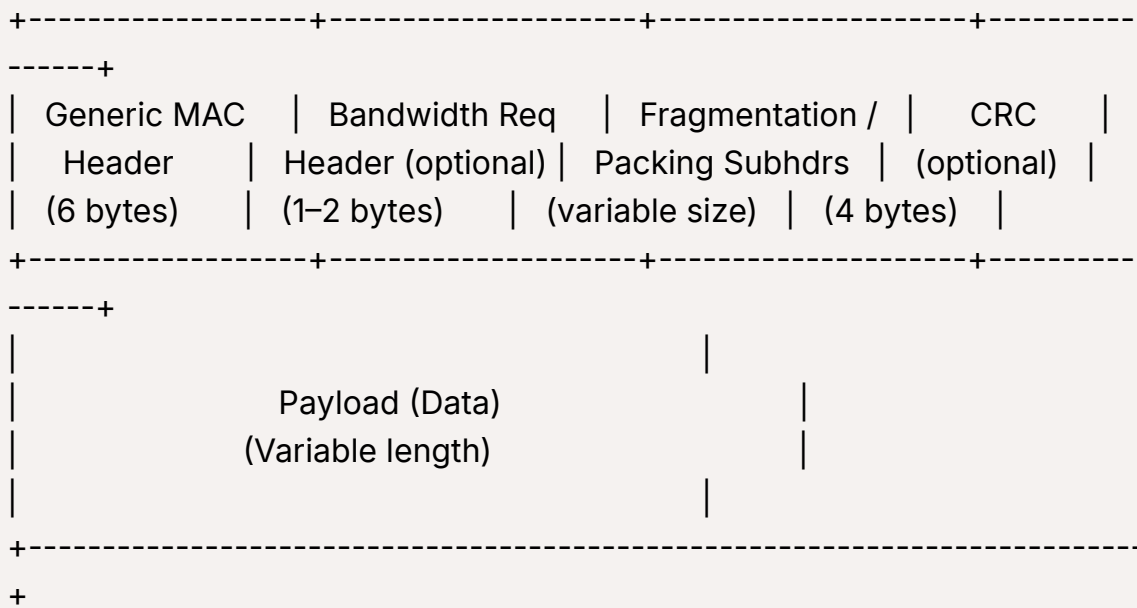


Fig.8 MAC PDU frame carrying several variable length MSDUs packed together

FSH → Fragmentation Sub-header

PSH → Packing Sub-header

The type of payload is identified by the sub-header immediately precedes it. For example FSH or PSH of above figure.



## Field Details:

### 1. Generic MAC Header (6 bytes)

- Always present.
- Contains:
  - HT (Header Type)
  - EC (Encryption Control)
  - Type (PDU type)
  - CI (CRC Indicator)
  - EKS (Encryption Key Seq.)
  - LEN (Length of PDU)
  - CID (Connection Identifier)
  - HCS (Header Check Sequence)

### 2. Bandwidth Request Header (1-2 bytes, optional)

- Used by subscriber stations (SS) to request bandwidth from the base station (BS).

### 3. Fragmentation / Packing Subheaders (Variable, optional)

- Used if a large SDU is split across PDUs (**fragmentation**)
  - Or multiple small SDUs are packed into one PDU (**packing**).
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#### 4. Payload (Variable length)

- The actual **data or management messages** carried.
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#### 5. CRC (4 bytes, optional)

- Used for **error detection** at the MAC PDU level.
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✅ **Quick Memory Tip for Exams:**

👉 Frame = **Header + [Optional Headers/Subheaders] + Payload + [CRC]**

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## 8. Explain the Femtocell deployment, OFDMA based physical layer, and MIMO of LTE.

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## LTE Advanced Concepts

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### 1 Femtocell Deployment



#### What is a Femtocell?

- A **small, low-power base station** used inside homes, offices, or small areas.
- Connected to the service provider through **broadband internet (DSL/Fiber)**.



#### Features:

- Improves **indoor coverage** (where macrocell signal is weak).
- Supports **few users (2–8 typical, sometimes up to 16–32)**.
- Reduces **load on macro network**.
- Saves **battery life** of mobile devices (stronger indoor signal).



#### Analogy:

Think of a femtocell as a **mini Wi-Fi router**, but instead of Wi-Fi, it gives you **cellular LTE coverage** at home.

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## 2 OFDMA-Based Physical Layer (LTE)

LTE uses **OFDMA** in the **downlink (base station → user)** because it supports many users efficiently.

### **Process (Step by Step):**

1. **Data** → Error correction coding + interleaving.
2. **Modulation** → QPSK, 16-QAM, or 64-QAM.
3. **Subcarrier Mapping** → Data placed on orthogonal subcarriers.
4. **IFFT** → Converts frequency subcarriers into time-domain OFDM signal.
5. **Cyclic Prefix** → Added to avoid multipath interference.
6. **Transmission** → Signal sent to users simultaneously, each using their own set of subcarriers.

### **Benefits:**

- Handles multipath fading well.
- High spectral efficiency.
- Supports multiple users at the same time.
- Flexible resource allocation.

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## 3 MIMO in LTE (Multiple Input Multiple Output)

### **What is MIMO?**

MIMO uses **multiple antennas** at both transmitter and receiver to improve **capacity, speed, and reliability**.

### **Types of MIMO in LTE:**

1. **Spatial Multiplexing**
  - Transmits **different data streams** on multiple antennas.
  - Increases **data rate** (parallel transmission).
2. **Transmit Diversity**

- Same data sent on multiple antennas with coding.
- Improves **reliability** in bad channel conditions.

### 3. Beamforming

- Antenna signals combined in a way to focus power towards the user.
- Improves **signal strength and reduces interference**.

## Quick Summary Table

| Feature  | Femtocell                      | OFDMA Physical Layer           | MIMO                        |
|----------|--------------------------------|--------------------------------|-----------------------------|
| Purpose  | Indoor coverage                | Efficient transmission         | High speed & reliability    |
| Key Idea | Small cell at home/office      | Subcarriers shared among users | Multiple antennas           |
| Benefit  | Better coverage, offload macro | Multipath handling, high data  | More data rate, less errors |

 **In Exam:** Write 1 short paragraph for each + a small diagram (Femtocell → small base station, OFDMA grid, MIMO antennas).

## 9. Draw the architecture of LTE and mention the combined function of S-GW and MME.

The architecture of LTE consists of three major parts:

- ✓ **User Equipment (UE),**
- ✓ **E-UTRAN (Evolved Universal Terrestrial Radio Access Network)**
- ✓ **EPC (Evolved Packet Core)**

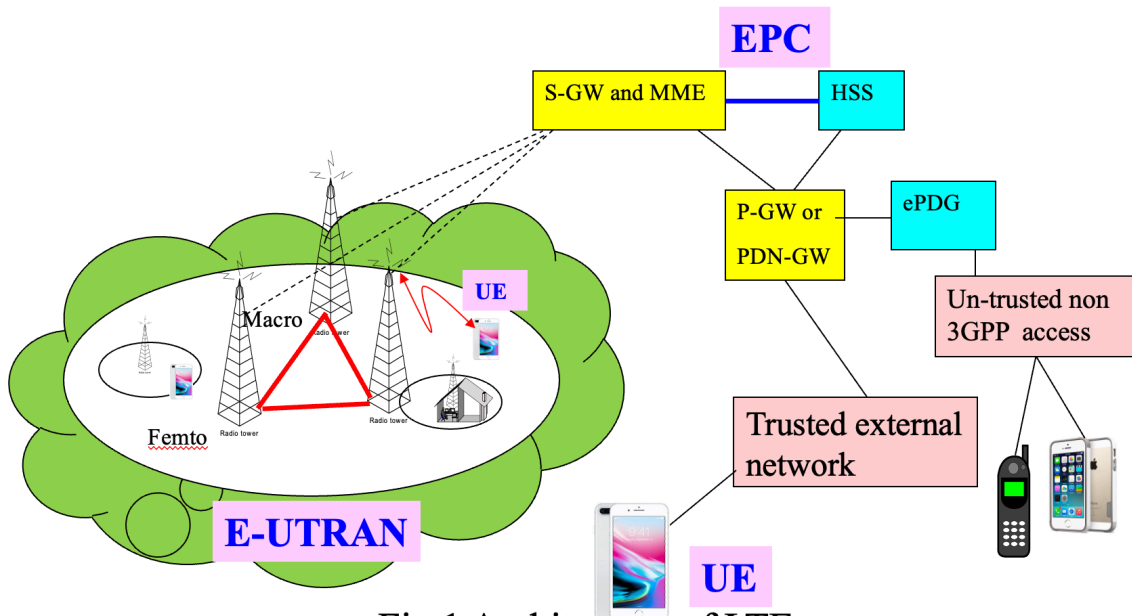
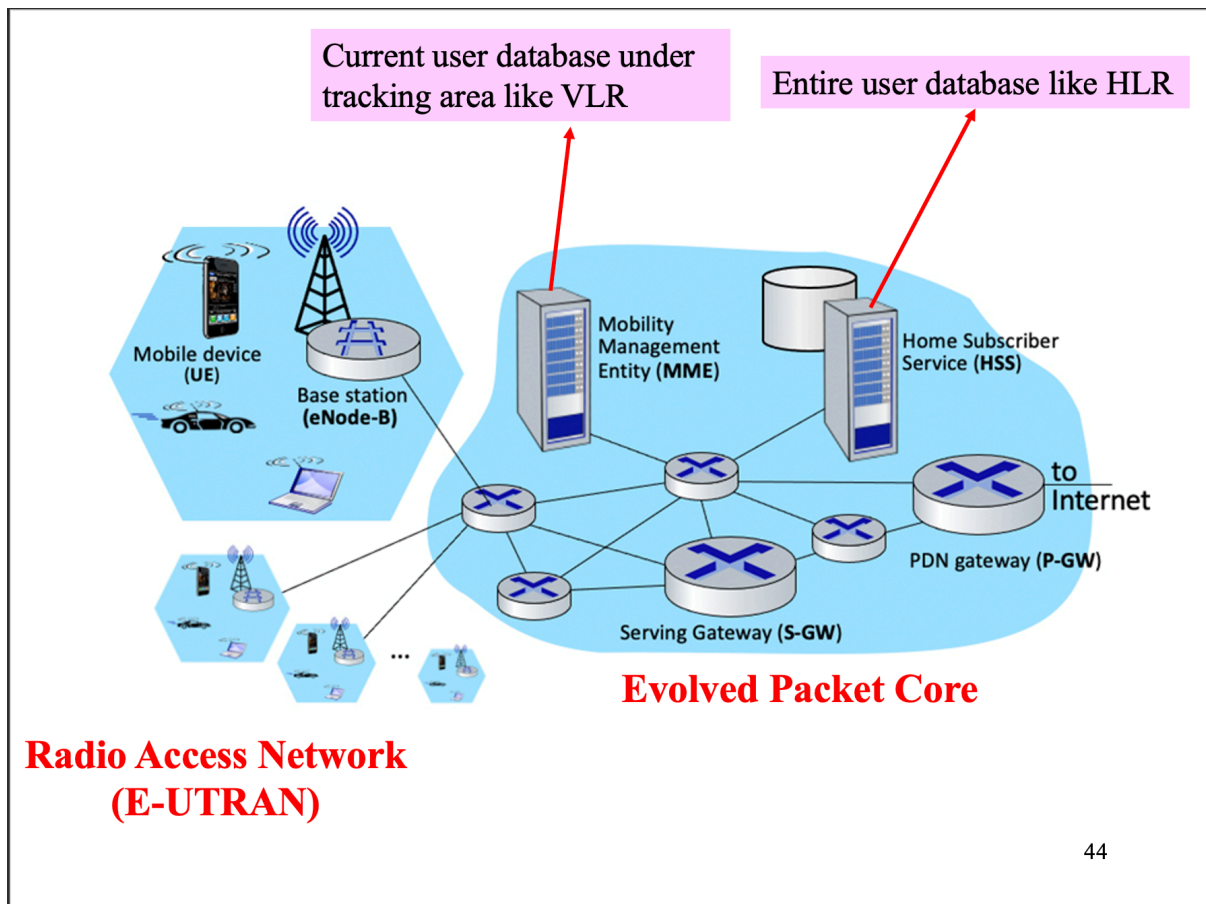


Fig.1 Architecture of LTE

45



44

## Key Elements

- **UE (User Equipment):** Smartphone, tablet, modem.
- **eNodeB (Evolved Node B):** Base station, handles radio communication.
- **MME (Mobility Management Entity):** Control plane node (authentication, mobility, signaling).
- **S-GW (Serving Gateway):** User plane node (data routing, mobility anchor).
- **P-GW (Packet Gateway):** Connects LTE to Internet/other networks.

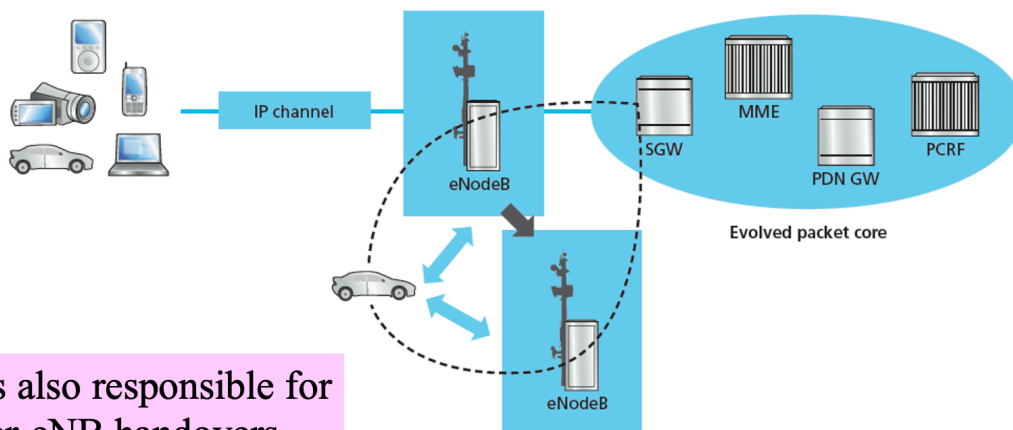
## Combined Function of S-GW and MME

Although separate in the EPC, the **S-GW (user plane)** and **MME (control plane)** work together tightly:

### S-GW Functions (User Plane):

### The serving gateway (S-GW)

- ✓ The S-GW connects **E-UTRAN** and **Evolved Packet Core (EPC)**.
- ✓ When terminals move across areas served by eNB elements in E-UTRAN, the S-GW serves as a **local mobility anchor**. This means that packets are routed through this point for intra E-UTRAN mobility (**within tracking area**) and mobility with other 3GPP technologies, such as 2G/GSM and 3G/UMTS.



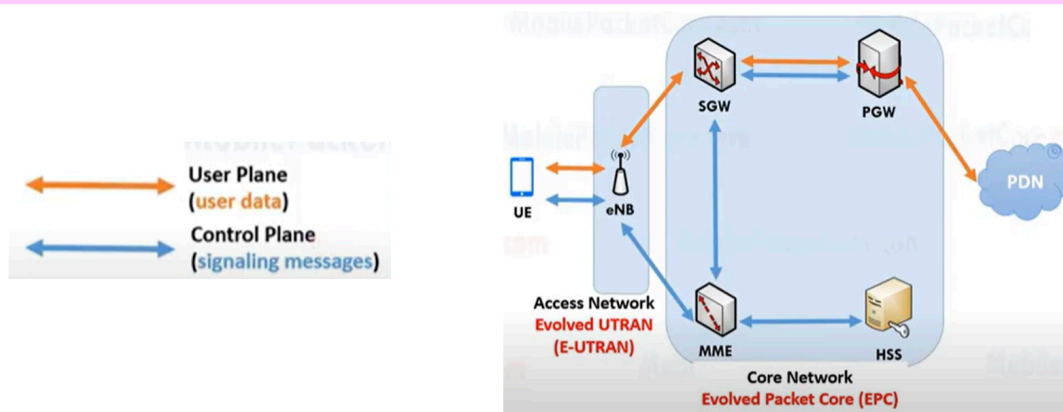
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- Routes and forwards user data packets.
- Acts as **mobility anchor** during handovers between eNodeBs.
- Buffers downlink data when UE is idle.

## ✓ MME Functions (Control Plane):

### MME (Mobility Management Entity)

✓ MME is responsible for initiating **paging and authentication** of the mobile device by interacting with the HSS (Home Subscriber Server). HSS has a database keeps subscription information of all users. When authentication is done then link is established between eNB and S-GW.



✓ MME retains location information at the **tracking area** level for each user, which is updated with roaming of a user (A large geographical region can be divided into multiple tracking areas and each **tracking area** may have hundred of cells).

- Authenticates users (with USIM/HSS).
- Manages mobility (tracking area updates, handovers).
- Establishes bearer paths for data.



### Combined Role (MME + S-GW together):

- Together they act like the **"brain and heart"** of LTE mobility:
  - **MME = brain** → controls signaling, user identity, session setup.



- **S-GW = heart** → forwards user data and keeps session alive across mobility.
- When a UE moves (handover), **MME coordinates** and **S-GW anchors the data path**, so the session remains uninterrupted.

#### **Analogy:**

- **MME = school principal** (decides rules, manages students).
- **S-GW = school bus service** (keeps students moving around smoothly).

👉 Together they ensure students (UEs) are both **authorized and continuously connected**.

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### **Quick Exam Recap**

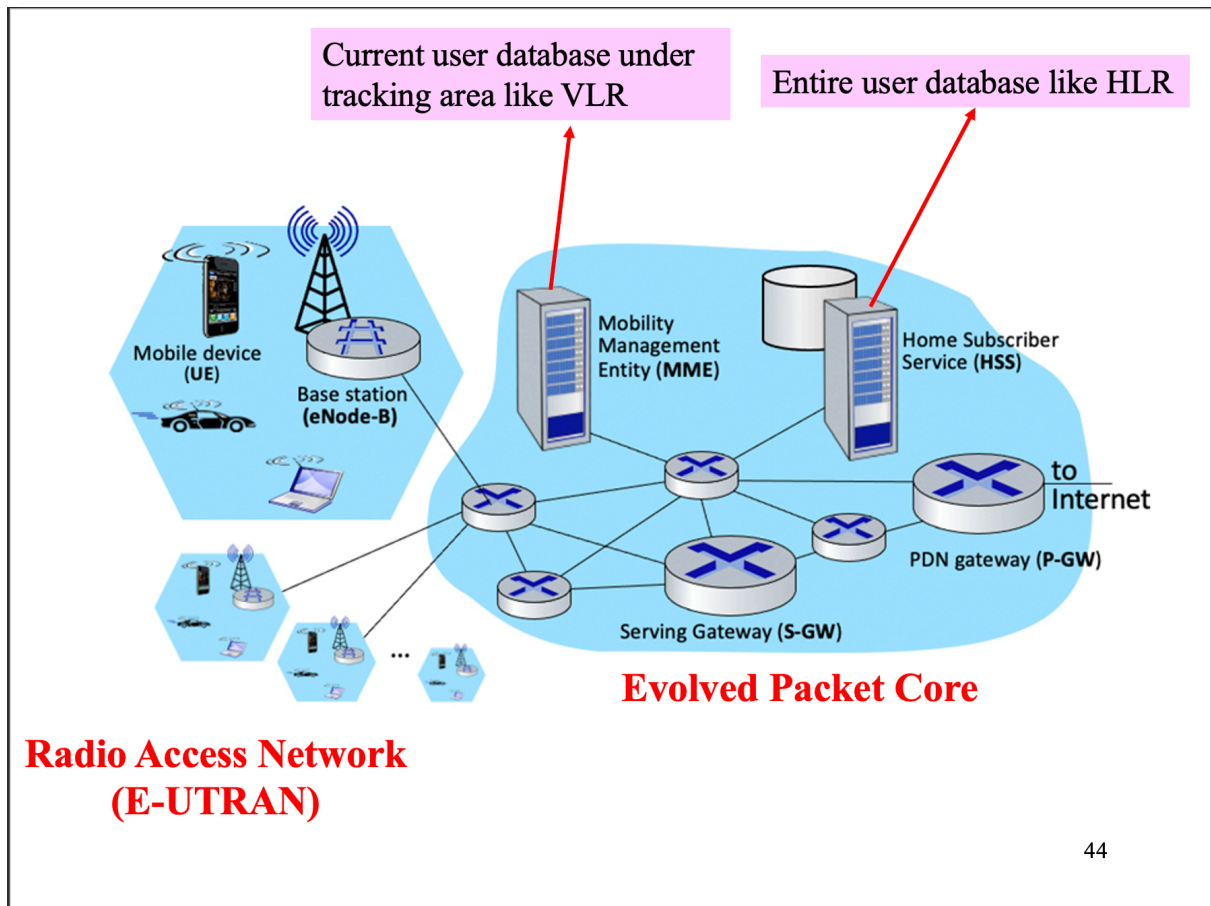
- **Architecture:** UE ↔ eNodeB ↔ EPC (MME + S-GW + P-GW).
  - **S-GW + MME together:** Handle mobility, session management, user authentication, and data anchoring.
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## 10. Explain Evolved UTRAN (E-UTRAN).

### **Evolved UTRAN (E-UTRAN)**

#### **What it is:**

- **E-UTRAN** is the **radio access network of LTE (Long Term Evolution)**.
- It is a **simplified version of UMTS (3G) UTRAN**, designed for **high data rates, low latency, and all-IP communication**.

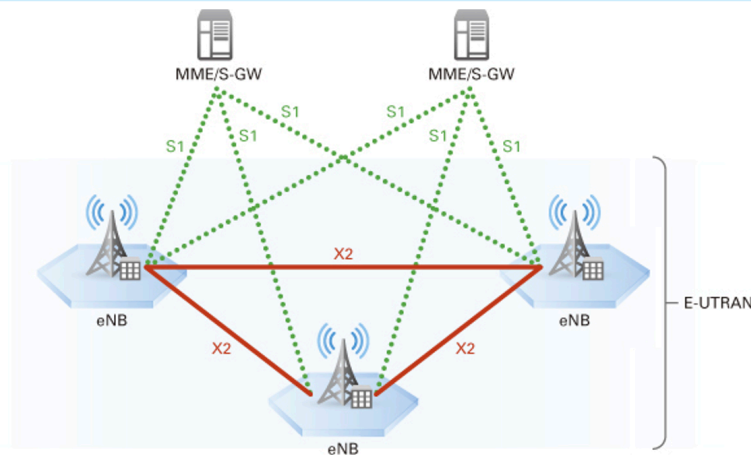


## Evolved UTRAN (E-UTRAN)

✓The E-UTRAN in LTE consists of **directly interconnected** eNBs which are connected to each other through the X2 interface and to the core network through the S1 interface.

✓This eliminates one of the biggest drawbacks of the former 3G systems (UMTS): the need to connect and control the NBs through the **Radio Network Controller (RNC)**, which make the system vulnerable to RNC failures.

The LTE E-UTRAN architecture can be seen in Figure below.



47

## Architecture of E-UTRAN

### 1. eNodeB (Evolved NodeB)

- The main network element in E-UTRAN.
- Works as both **base station + part of the controller**.
- Connects directly to user devices (UE – User Equipment).

### 2. No RNC (Radio Network Controller)

- In 3G UTRAN, RNC controlled the NodeBs.
- In LTE, RNC is removed → eNodeB takes over RNC functions.
- This reduces complexity and latency.

### 3. Interfaces:

- **Uu interface** → Between UE and eNodeB (air interface).
- **X2 interface** → Between eNodeBs (for handover & coordination).
- **S1 interface** → Between eNodeB and EPC (Evolved Packet Core).

## 🌟 Key Features of E-UTRAN

- **Flat architecture** → Only eNodeBs, no RNC.
  - **High spectral efficiency** (uses OFDMA for downlink, SC-FDMA for uplink).
  - **Low latency** → Important for real-time applications (gaming, VoIP).
  - **Flexible bandwidths** (1.4 MHz to 20 MHz).
  - **Supports MIMO** for higher capacity and reliability.
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## 🧠 Simple Analogy:

Think of 3G UTRAN as a **school system** where teachers (NodeBs) must always ask the principal (RNC) before doing anything.

In LTE's E-UTRAN, each teacher (eNodeB) is **independent** — they teach and manage directly, making the system **faster and simpler**.

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## ✅ Quick Exam Recap

- E-UTRAN = LTE Radio Access Network.
  - Main element = eNodeB.
  - Interfaces = Uu, X2, S1.
  - Advantages = flat, fast, efficient, supports MIMO & OFDMA.
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## 11. What are the basic function of User Equipment (UE) and USIM.

## 📱 User Equipment (UE) and USIM Functions in LTE

### 1 User Equipment (UE)

UE = **the mobile device used by the subscriber** (smartphone, tablet, modem).

UE refers the the devices that is used by the end user like smartphone, tablet, modem.

It consists of:

- **Mobile Terminal (MT)** – the hardware (radio, processor, display, etc.).
- **USIM (Universal Subscriber Identity Module)** – the SIM card that stores subscriber info.

## **Basic Functions of UE:**

### 1. **Radio Communication**

- Sends/receives signals to/from the network via eNodeB.

### 2. **Mobility Management**

- Handles cell selection, reselection, and handover while moving.

### 3. **Authentication**

- Works with USIM to prove the user's identity to the network.

### 4. **Session Management**

- Establishes and maintains data/voice sessions with the network.

### 5. **QoS Handling**

- Requests and uses resources as per Quality of Service (e.g., video call vs. browsing).

### 6. **User Interface**

- Provides the actual service to the user (calls, internet, SMS, apps).

---

## **USIM (Universal Subscriber Identity Module)**

USIM = The **smart card (SIM)** inside UE that stores subscriber info securely.

## **Basic Functions of USIM:**

### 1. **User Identity**

- Stores **IMSI (International Mobile Subscriber Identity)** to identify the subscriber.

### 2. **Authentication & Security**

- Provides secret keys for authentication with the network.
- Supports encryption and integrity protection (secure communication).

### 3. Subscriber Information Storage

- Stores phone number, service subscription info, and temporary identities.

### 4. Mobility Management Support

- Helps in tracking location (Location Area Identity, Temporary IDs).

### 5. Application Support

- Can store additional applications (like mobile banking, operator apps).
- 

## Simple Analogy:

- **UE = The smartphone (the whole device)** – does the “work” of connecting, transmitting, and running apps.
  - **USIM = The ID card inside it** – tells the network **who you are** and makes sure only you can access your subscribed services.
- 

## Quick Recap (Exam-Friendly):

- **UE Functions:** Communication, Mobility, Authentication, Session Mgmt, QoS, User Interface.
  - **USIM Functions:** Identity, Authentication/Security, Subscriber info, Mobility support, Applications.
- 

## 12. Write in short about the working principle of P-GW, ePDG and PCRF.

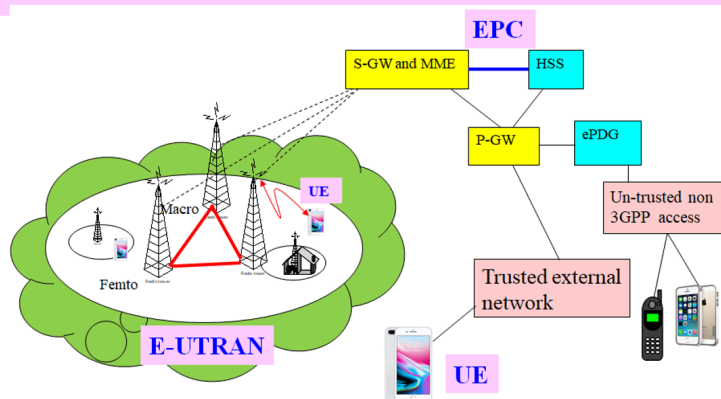
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## Working Principles

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## Packet Data Network Gateway (P-GW)

- ✓ Like the S-GW, the **Packet Data Network Gateway (PDN GW)** is the termination point of the packet data interface towards the Packet Data Network(s). As an anchor point for sessions towards the external Packet Data Networks.
- ✓ **ePDG** (Evolved Packet Data Gateway) provides secured data transmission between UE to un-trusted non-3GPP access through EPC.



52

### 1 P-GW (Packet Data Network Gateway)

- Acts as the **exit/entry point** between LTE network and external packet data networks (like the Internet, IMS, corporate intranets).
- **Functions:**
  - Assigns **IP address** to the UE.
  - Handles **policy enforcement** (QoS, charging rules).
  - Provides **packet filtering** and routing.
- **Analogy:** Like the **main gateway/router** that connects your phone to the Internet.

### 2 ePDG (Evolved Packet Data Gateway)

- Used for **secure access** when UE connects to the LTE core over **untrusted non-3GPP networks** (e.g., public Wi-Fi).
- **Functions:**

- Establishes secure **IPsec tunnel** between UE and core network.
- Ensures authentication & encryption of data.
- **Analogy:** Like a **security guard + VPN gateway**, making sure that even if you connect through coffee shop Wi-Fi, your LTE data is safe.

### 3 PCRF (Policy and Charging Rules Function)

- A **centralized policy server** in EPC.
- **Functions:**
  - Decides QoS (Quality of Service) for each user session.
  - Applies charging rules (e.g., different data plan rules, prepaid vs postpaid).
  - Communicates with P-GW to enforce these policies.
- **Analogy:** Like the **traffic manager + billing officer** → decides who gets priority lanes and how much to charge them.

### ✓ Quick Recap Table

| Component   | Role in LTE                                        | Simple Analogy                    |
|-------------|----------------------------------------------------|-----------------------------------|
| <b>P-GW</b> | Connects LTE to Internet, assigns IP, enforces QoS | Main router/gateway               |
| <b>ePDG</b> | Provides secure access via untrusted Wi-Fi         | VPN security guard                |
| <b>PCRF</b> | Manages QoS and charging rules                     | Traffic manager + billing officer |

## 13. Draw the LTE-A protocol stack. Mention the function of PDCP and RLC.

### LTE-A Protocol Stack

LTE protocol stack has **two planes**:

- **User Plane** → carries user data (e.g. Internet traffic).



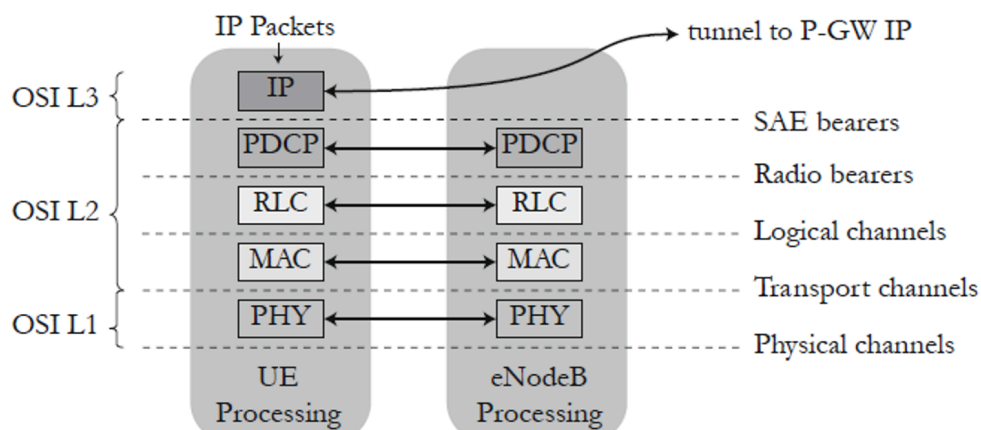
- **Control Plane** → carries signaling (RRC, NAS messages).



## Text Diagram (Simplified View)

✓PDCP layer or layer 2 **Packet Data Convergence Protocol** is responsible for data ciphering and IP header compression to reduce the IP header overhead.

✓RLC layer or layer 2 **Radio Link Control** performs the data concatenation and then generates the segmentation of packets from IP-Packets of random sizes which comprise a Transport Block (TB) of size adapted to the radio transfer. The RLC layer handles a retransmission scheme of lost data through a first level of **Automatic Repeat reQuests** (ARQ).



30



## Functions of PDCP (Packet Data Convergence Protocol)

- **Header Compression:** Reduces IP/TCP/UDP headers to save bandwidth.
- **Security:** Provides ciphering (encryption) and integrity protection.
- **In-sequence Delivery:** Ensures packets are delivered in correct order after handover.
- **Duplicate Removal:** Avoids forwarding duplicates during handovers.



Think of PDCP as the “security guard + data compressor”.

## **Functions of RLC (Radio Link Control)**

- **Segmentation & Reassembly:** Splits large packets into smaller ones, reassembles them at receiver.
- **Error Correction (ARQ):** Retransmits lost packets.
- **Flow Control:** Prevents buffer overflow.
- **Modes:**
  - AM (Acknowledged Mode – reliable with retransmission)
  - UM (Unacknowledged Mode – faster, no retransmission, e.g. video streaming)
  - TM (Transparent Mode – just pass data, e.g. control signaling).

 **Think of RLC as the “traffic manager + packet repairman”.**

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## **Quick Recap for Exams**

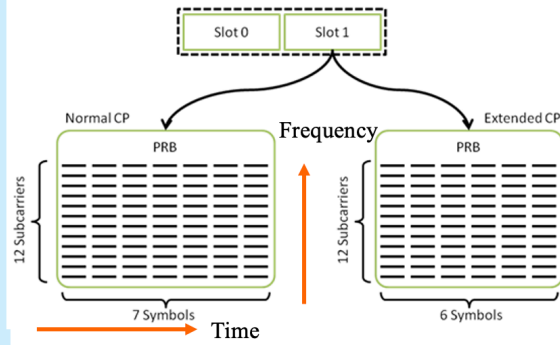
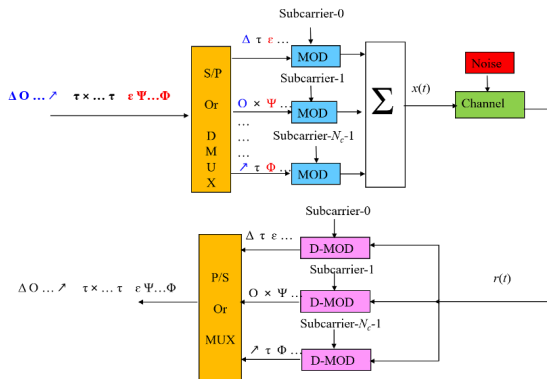
- **Protocol stack layers (top → bottom):** NAS → RRC → PDCP → RLC → MAC → PHY.
- **PDCP:** Compression, encryption, integrity, ordering.
- **RLC:** Segmentation, ARQ, flow control, error recovery.

**14. Draw the Physical Resource Blocks (PRB) under both Normal CP and Extended CP of LTE-A. If the separation is 15 KHz then determine the length of the slot of PRB for both the cases.**

## **Physical Resource Block (PRB) in LTE-A**

- **PRB bandwidth:** 12 subcarriers × 15 kHz = **180 kHz**
  - **Time span per PRB:** 1 slot = **0.5 ms** (for both Normal CP and Extended CP)
  - Difference is the **number of OFDM symbols per slot**.
-

✓ Each slot contains either six or seven OFDM symbols, depending on the **Cyclic Prefix (CP)** length. In LTE-A the separation between subcarrier  $\Delta f = 15\text{KHz}$ . The useful symbol time is  $1/15\text{ kHz} = 66.6\text{ microsec}$ . Since **normal CP** is about 4.69 microsec long, seven OFDM symbols can be placed in the 0.5-ms slot as each symbol occupies  $(66.6 + 4.69) = 71.29\text{ microseconds}$ .

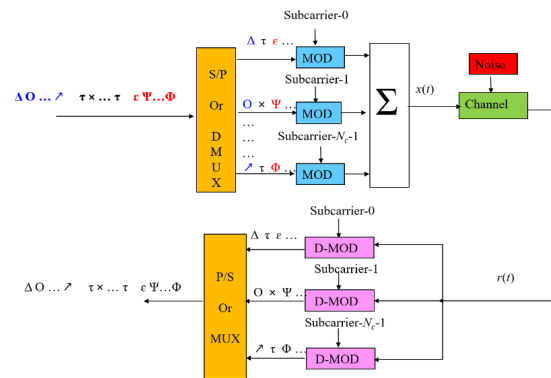
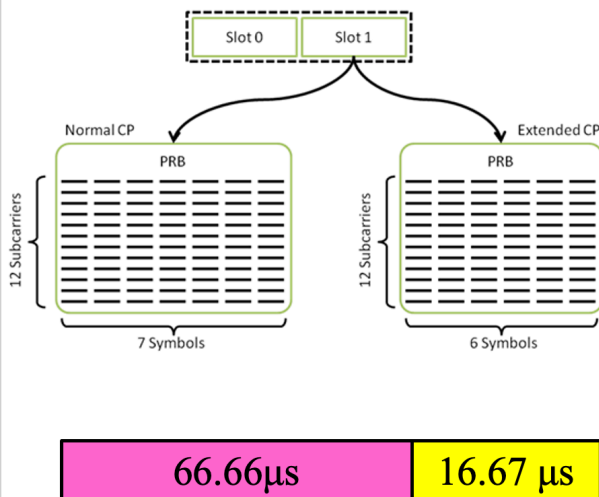


**Physical Resource Blocks (PRB)**

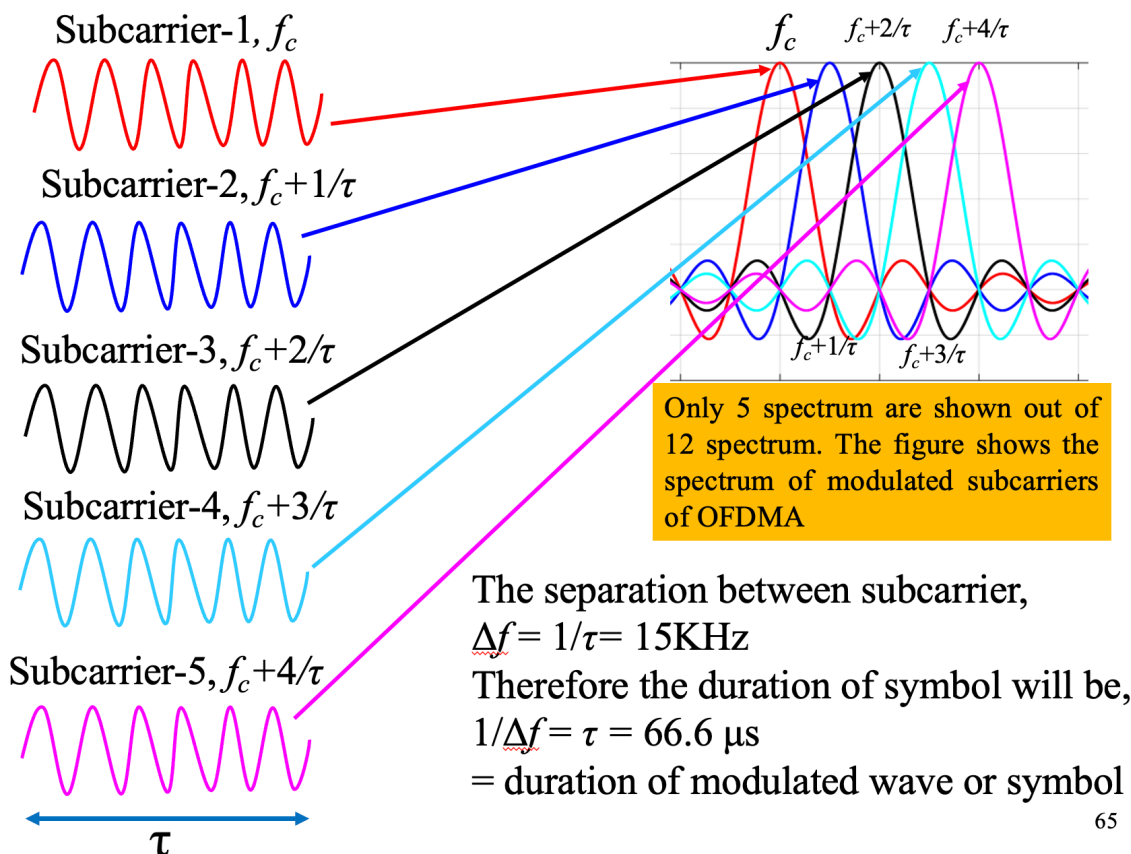


Now,  $71.29 \times 7 = 500\text{ microseconds} = 0.5\text{ms} = \text{Length of slot}$

✓ When **extended CP** (=16.67 microsec) is used the total OFDM symbol time is  $(66.6 + 16.67) = 83.27$  microseconds. Six OFDM symbols can then be placed in the 0.5-ms slot. Now,  $83.27 \times 6 = 500$  microseconds = 0.5ms = Length of slot



64



65

---

✅ **Final:**

**Slot length = 0.5 ms for both Normal CP and Extended CP**

(They differ in **symbols/slot**: 7 vs 6, not in slot duration.)

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## 15. Draw LTE FDD frame structure.

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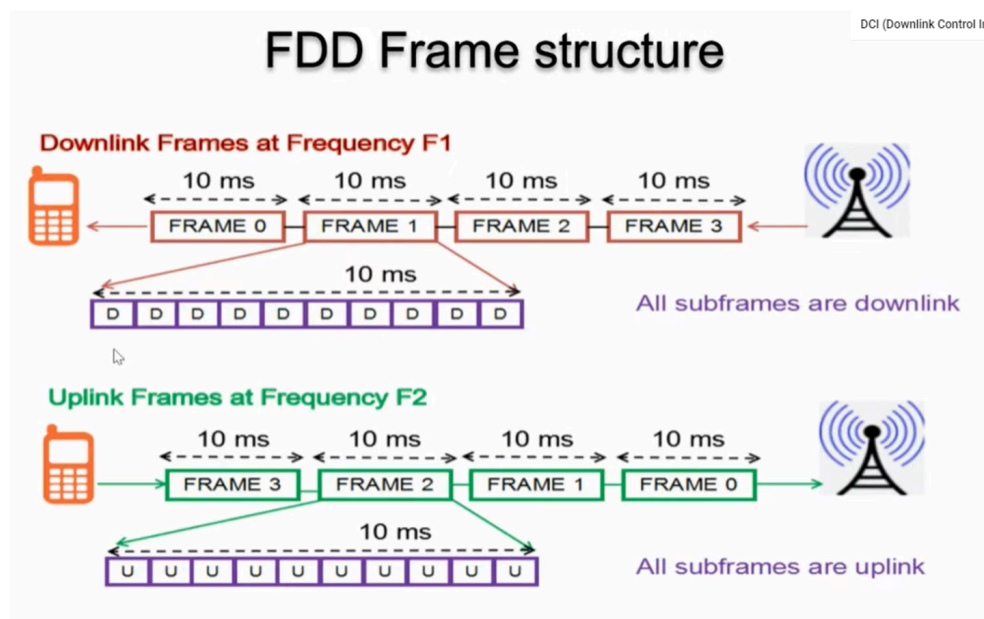
### **LTE FDD Frame Structure**

👉 In FDD (Frequency Division Duplex):

- **Downlink (DL)** and **Uplink (UL)** use **separate frequency bands**.
  - Both transmit **at the same time** (parallel transmission).
  - Each **radio frame = 10 ms**, divided into **10 subframes** (1 ms each).
- 



### **Text Diagram of LTE FDD Frame**



60

### Legend:

- **Radio Frame** = 10 ms.
- **Each frame** = 10 subframes (SF0–SF9), 1 ms each.
- DL and UL happen **simultaneously** but on **different frequencies**.
- No special subframe (unlike TDD).

### Key Points for Exam:

- LTE FDD uses **paired spectrum** (separate UL & DL bands).
- **Both directions are continuous** → no switching needed.
- Lower latency than TDD because UL/DL are simultaneous.

### Analogy:

Think of a **two-way road**  —

- One lane = **Downlink traffic**
- Other lane = **Uplink traffic**

Both directions flow **at the same time without colliding**, because each has its own lane (frequency).

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## 16. Draw LTE TDD frame structure.



### **LTE TDD Frame Structure**

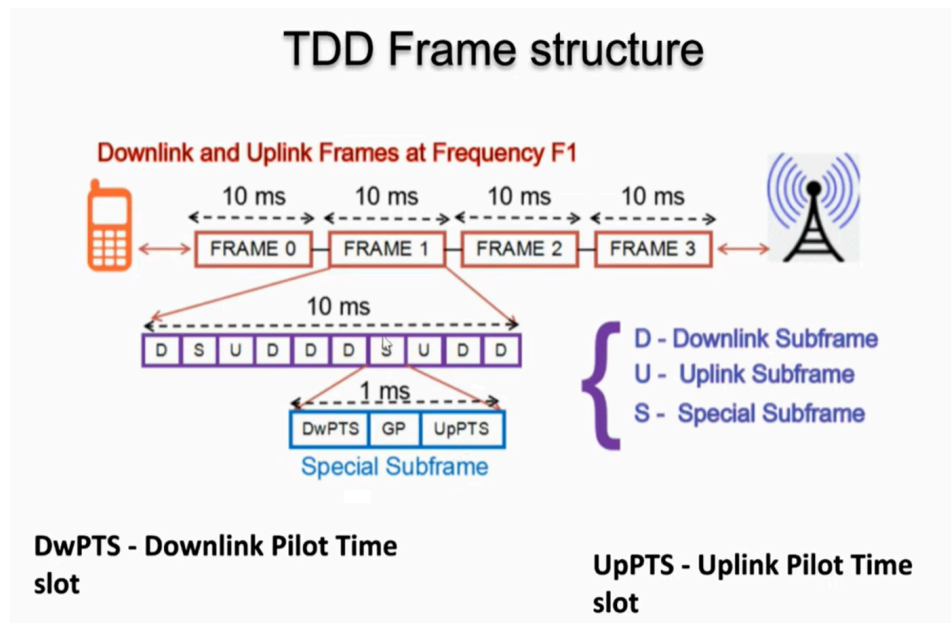
👉 In **TDD (Time Division Duplex)**, uplink (UL) and downlink (DL) share the **same frequency band** but are separated in **time slots**.

- One LTE **radio frame** = **10 ms**
  - Divided into **10 subframes (each 1 ms)**
  - Some subframes = **Downlink**, some = **Uplink**, some = **Special subframe (DwPTS, GP, UpPTS)**
- 



### **Text Diagram of LTE TDD Frame (Example: Config 1)**

## TDD Frame structure



62

### Legend:

- **DL = Downlink subframe** (eNodeB → UE)
- **UL = Uplink subframe** (UE → eNodeB)
- **S = Special Subframe** (split into 3 parts):
  - **DwPTS (Downlink Pilot Time Slot)** → for DL sync/control
  - **GP (Guard Period)** → avoids UL/DL overlap
  - **UpPTS (Uplink Pilot Time Slot)** → for UL sync

### Key Points for Exam:

- LTE TDD frame = 10 ms, 10 subframes.
- DL and UL share same band, separated in time.
- Special subframe (S) contains **DwPTS + GP + UpPTS**.



- Different **TDD configurations** exist (more DL-heavy or UL-heavy depending on traffic).

---

### Simple Analogy:

Think of a **one-way bridge**  — cars (data) go **downlink** for some time, then the bridge is cleared (special subframe), and then cars go **uplink**.

---

## 17. Give the basic concept of CP. Why it is used in OFDMA.



## Cyclic Prefix (CP) in OFDMA



### Basic Concept of CP

- A **Cyclic Prefix (CP)** is a **copy of the end part of an OFDM symbol**, which is placed at the beginning of the symbol before transmission.
- It is **not new data**, just a repetition, and acts like a **guard interval** between consecutive symbols.



### Why CP is Used in OFDMA?

#### 1. Removes Inter-Symbol Interference (ISI)

- In wireless, signals reflect off buildings/objects → multipath propagation.
- Reflected signals from the previous symbol may overlap with the next symbol → ISI.
- CP provides a “buffer zone” so these reflections die out before the useful data starts.

---

#### 2. Preserves Orthogonality of Subcarriers

- OFDMA relies on subcarriers being orthogonal.
  - Multipath can break this property and cause interference.
  - CP ensures subcarriers remain orthogonal, avoiding Inter-Carrier Interference (ICI).
-

### 3. Simplifies Equalization

- With CP, the wireless channel looks like a **circular convolution**.
  - This makes it easy to correct channel effects using **FFT at the receiver**.
- 

### 🌟 Key Points to Write in Exam:

- CP = copy of last part of symbol, added in front.
  - Used to handle multipath effects.
  - Eliminates ISI, preserves orthogonality, and simplifies receiver design.
- 

### 🧠 Simple Analogy:

Imagine people walking through a door 🚪 one after another:

- If the next person starts walking **too early**, they bump into the previous one (interference).
  - CP works like a **buffer space** between people → ensures smooth entry without collision.
- 

### ✅ Quick Recap

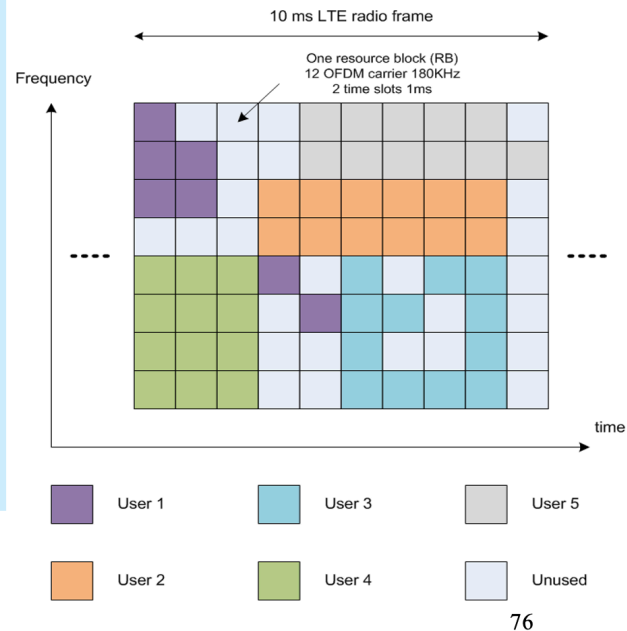
- **Concept:** Copy of end of OFDM symbol added at start.
  - **Purpose:** Remove ISI, maintain orthogonality, ease equalization.
- 

## 18. Explain OFDMA time frequency scheduling.

### 📡 OFDMA Time–Frequency Scheduling

# OFDMA time-frequency scheduling

- ✓ Minimum allocate-able resource in LTE is **Resource Block pair**
- ✓ Resource block pair is 12 subcarriers wide in frequency domain and lasts for two time slots (1ms)
- ✓ Depending on the length of cyclic prefix RB pair may have 14 or 12 OFDM symbols
- ✓ PHY channels consist of certain number of allocated RB pairs
- ✓ Overhead channels are typically in a predetermined location in time frequency domain
- ✓ Allocation of the radio block is done by scheduler at eNode B



## Basic Idea

In **OFDMA**, the available radio resources are divided into:

- **Time domain** → different time slots (subframes).
- **Frequency domain** → different subcarriers (grouped into Resource Blocks, RBs).

👉 Together, they form a **time-frequency grid**.

The **scheduler** (at eNodeB in LTE / BS in WiMAX) decides **which user gets which resource blocks in which time slot**.

## How Scheduling Works (Step by Step):

### 1. Resource Grid Formation

- Divide bandwidth into **RBs (Resource Blocks)** in frequency.
- Divide transmission time into **TTIs (Transmission Time Intervals)** in time.

## 2. User Demands & Channel Quality

- Scheduler checks **who needs data** and **how good/bad their channel is** (via CQI – Channel Quality Indicator).

## 3. Allocation

- Assigns RBs dynamically to users based on:
  - QoS requirements (voice, video, data).
  - Fairness (not always giving best resources to one user).
  - Efficiency (maximizing throughput).

## 4. Types of Scheduling

- **Time-domain scheduling** → each user gets full bandwidth for a short time.
- **Frequency-domain scheduling** → each user gets specific RBs across frequency.
- Often a **mix of both** is used.

---

## ✨ Benefits of Time–Frequency Scheduling in OFDMA:

- Supports **multiuser access** efficiently.
- Provides **flexibility** to adapt resources.
- Improves **spectral efficiency**.
- Meets **QoS requirements** for different apps (e.g., VoIP vs. video).
- Reduces **interference** by careful RB allocation.

---

## **Analogy:**

Think of a **classroom timetable**:

- **Time slots = lecture periods**
- **Frequency blocks = classrooms**
- The scheduler (teacher) assigns **which student (user) gets which classroom at which period**.

👉 This way, all students get fair usage of resources.

---

## ✅ Quick Recap

- **OFDMA Scheduling** = assigning time + frequency resources to users.
  - Based on user demand, channel quality, and QoS.
  - Ensures efficiency, fairness, and high throughput.
- 

## 19. Explain uplink and downlink reference signal.

---



## Reference Signals in LTE

Reference signals (RS) are **special known signals** inserted into LTE frames to help the system with **channel estimation, synchronization, and measurements**.

They are crucial for reliable communication.

---

### 1 Downlink Reference Signals (DL RS)

- Transmitted by **eNodeB (base station)** to the UE.
- Purpose: help the **UE estimate(অনুমান) the downlink channel quality**.

### ✅ Types of DL RS:

#### 1. Cell-specific Reference Signal (CRS):

- Sent for all UEs in a cell.
- Used for channel estimation, demodulation, and handover measurements.

#### 2. UE-specific Reference Signal (UE-RS):

- Dedicated to a specific UE.
- Used mainly in **beamforming/MIMO** to optimize transmission for that user.

#### 3. Positioning Reference Signal (PRS):

- Used for **location services**.

## **Functions of DL RS:**

- Channel estimation for coherent demodulation.
  - Handover measurements (RSRP, RSRQ).
  - Support for MIMO/beamforming.
- 

## **2 Uplink Reference Signals (UL RS)**

- **Transmitted by UE** to the eNodeB.
- Purpose: help the **eNodeB estimate the uplink channel quality**.

## **Types of UL RS:**

### 1. **Demodulation Reference Signal (DM-RS):**

- Used for demodulating uplink data (PUSCH, PUCCH).

### 2. **Sounding Reference Signal (SRS):**

- Sent across the bandwidth to give eNodeB an idea of channel conditions over frequency.
- Helps eNodeB in **scheduling and resource allocation**.

## **Functions of UL RS:**

- Channel estimation for uplink demodulation.
  - Uplink scheduling (resource allocation by eNodeB).
  - Support for MIMO precoding.
- 

## **Simple Analogy:**

- **Downlink RS:** Like a teacher (eNodeB) sending a **sample question** to students (UEs) → so they know the pattern and can adjust.
  - **Uplink RS:** Like students (UEs) sending a **sample answer** → so the teacher knows how good/bad their condition is and can grade accordingly.
- 

## **Quick Recap (Exam-Ready)**

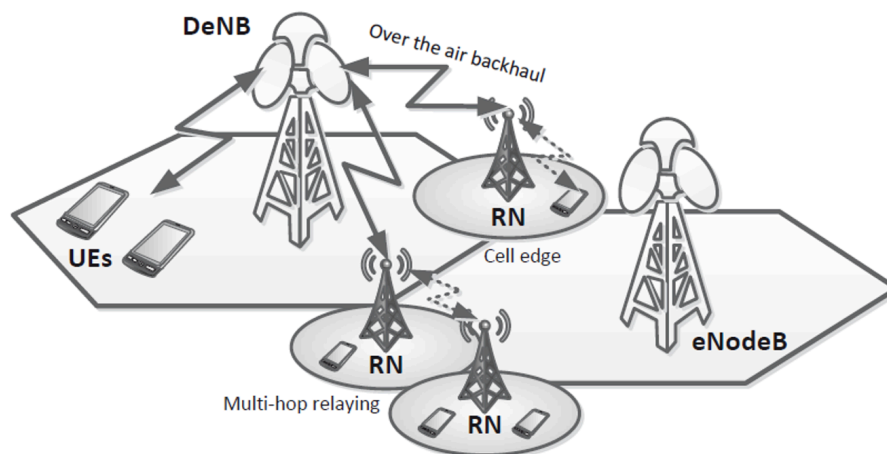
| Reference Signal               | Sent By     | Purpose                                                                |
|--------------------------------|-------------|------------------------------------------------------------------------|
| <b>DL RS</b> (CRS, UE-RS, PRS) | eNodeB → UE | Helps UE estimate channel, measure RSRP/RSRQ, support MIMO & handover  |
| <b>UL RS</b> (DM-RS, SRS)      | UE → eNodeB | Helps eNodeB estimate uplink channel, allocate resources, support MIMO |

## 20. Describe in short about Relays and Repeater.

### Relays vs Repeaters in Wireless Networks

#### Relays and Repeaters

Relaying is used in order to deploy cells in areas where no (or very expensive) wired backhaul exists. It is often used to improve the coverage and throughput, for example, in urban, and rural scenarios. Figure below shows a typical scenario where a Relay Node (RN) is connected to an eNodeB wirelessly. The eNodeB that the RN is connected to is called in that case, a Donor eNodeB (DeNB).



83

#### ◆ Repeater

- A **basic device** that simply **receives a weak signal and amplifies it**, then retransmits it.

- It does **not process** or decode the signal.
- Used mainly to **extend coverage** in areas with weak signals (like tunnels, basements).

#### ✅ Key Points:

- Simple analog device.
- Amplifies both signal **and noise**.
- Cheap and easy to deploy.

🧠 **Analogy:** Like a **loudspeaker** – it just makes the sound louder without changing it.

## ◆ Relay

- A **smarter device** than a repeater.
- It **receives, decodes, processes, and re-transmits** the signal.
- Supports **protocol-level functions** (like error correction, scheduling).
- Used in LTE/4G to extend coverage in **rural areas, indoors, or cell edges**.

#### ✅ Key Points:

- Digital device with processing power.
- Removes noise before retransmission (cleaner signal).
- More expensive than repeaters but gives better performance.

🧠 **Analogy:** Like a **translator** – listens carefully, understands, and then repeats in a clear voice.

## ✅ Quick Recap Table

| Feature        | Repeater                     | Relay                           |
|----------------|------------------------------|---------------------------------|
| Processing     | No (just amplifies)          | Yes (decodes + re-encodes)      |
| Signal Quality | Amplifies noise too          | Removes noise, cleaner signal   |
| Cost           | Cheaper                      | More expensive                  |
| Use Case       | Small coverage gaps (indoor) | Rural/edge coverage, LTE relays |